One responsibility.
One state owner.
Many possible backends.

Omega Hive × ASI:chain

From independent module contracts to a real Hive caller that recovers

M1.2 COMPLETE

M2 NOT STARTED

Architecture review • 16 October 2026 • verified native integration

The architecture separates three commitments

A function boundary, an authority boundary, and an evidence boundary.

01

Replace behavior safely

Select a versioned contract and semantic profile. Keep the consumer independent of the provider.

02

Write through the owner

Episodes, beliefs, attention and jobs each have a single logical authority for their state.

03

Name what was observed

Storage, delivery, execution and finality carry separate receipts and explicit limits.

One real Hive operation exercises these boundaries

STORE → SHARE REF → DELEGATE → RECOVER → ACCEPT → ACK

Fifteen modules; four responsibility groups

Real Hive consumes these ports; module, process and repository boundaries remain independent.

CLIENT / COGNITION

- Episodic memory
- Belief revision
- PLN kernel
- ECAN / attention
- Inference control

SHARED APPLICATION PORTS

- AtomStorage
- Relations · spec
- Search projection · spec
- Agent messaging
- Delegated execution

CONTROL & POLICY

- Registry
- Compatibility resolver
- Authorization

Local harness policy and static descriptors

PROVIDER ADAPTERS

DA storage

Real DeTTa libraries

Consensus / finality

M1 historical; M1.2 NOT_RUN

SPECIFICATION ONLY · IMPLEMENTED HARNESS · REAL PROVIDER INTEGRATION

Data responsibilities: bytes → references → history

The first five modules separate persistence, identity, structure, discovery and provenance.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
DA storage	Persist opaque bytes put · get · availability	Payloads and shares REAL LOCAL PROVIDER
AtomStorage	Assign scoped immutable references create · get · getMany · availability	Reference + replay metadata IMPLEMENTED
Relations	Represent ordered structural roles assertRelation · getRelation	Immutable relation records SPECIFIED
Search projection	Find references at a known watermark query	Rebuildable index SPECIFIED
Episodic memory	Preserve ordered events + provenance recordEpisode · appendEvent	Private episode history SPECIFIED · M2

Capture and retrieve data before choosing its meaning.

Cognitive responsibilities: history ≠ truth ≠ attention

All four modules below are typed specifications. Fusion adds no cognitive provider.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / EXCLUSION
Belief revision	Revise under an explicit numerical model submitEvidence · readBelief	Belief + evidence ledger No agenda or attention writes
PLN kernel	Apply a rule to explicit premises profiles · applyRule	No mutable domain state Returns a proposed derivation
ECAN / attention	Allocate, select and decay attention submitStimulus · select · advance	Attention balances + logical time No PLN dependency
Inference control	Schedule bounded reasoning work step	Step receipts + checkpoint No ownership of truth or attention

Mathematics and attention policy remain explicit review gates.

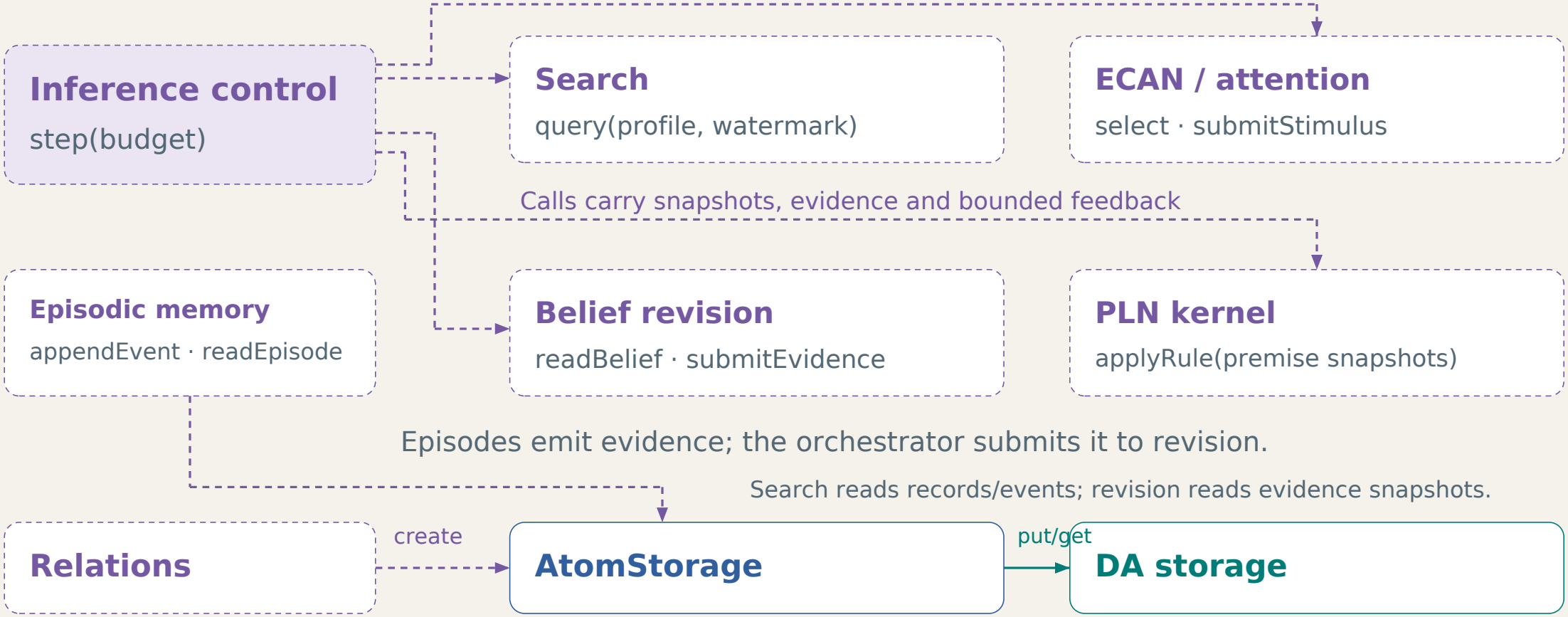
Control responsibilities: move work without moving authority

The remaining six modules connect owners and report exactly what happened.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
Messaging	Accept, deliver, acknowledge send · poll · ack	Inbox + deduplication · IMPLEMENTED
Execution	Run one named operation submit · status · result	Jobs + completion outbox · IMPLEMENTED
Registry	Describe configured providers descriptors	Static configuration · IMPLEMENTED
Resolver	Match required semantics resolve	No mutable state · IMPLEMENTED
Consensus / finality	Submit approved commitment; observe submit · observe	Signed intents · HISTORICAL M1
Authorization	Check current caller + scope + method authorize / live policy checks	Server-owned policy · LOCAL FIXTURE

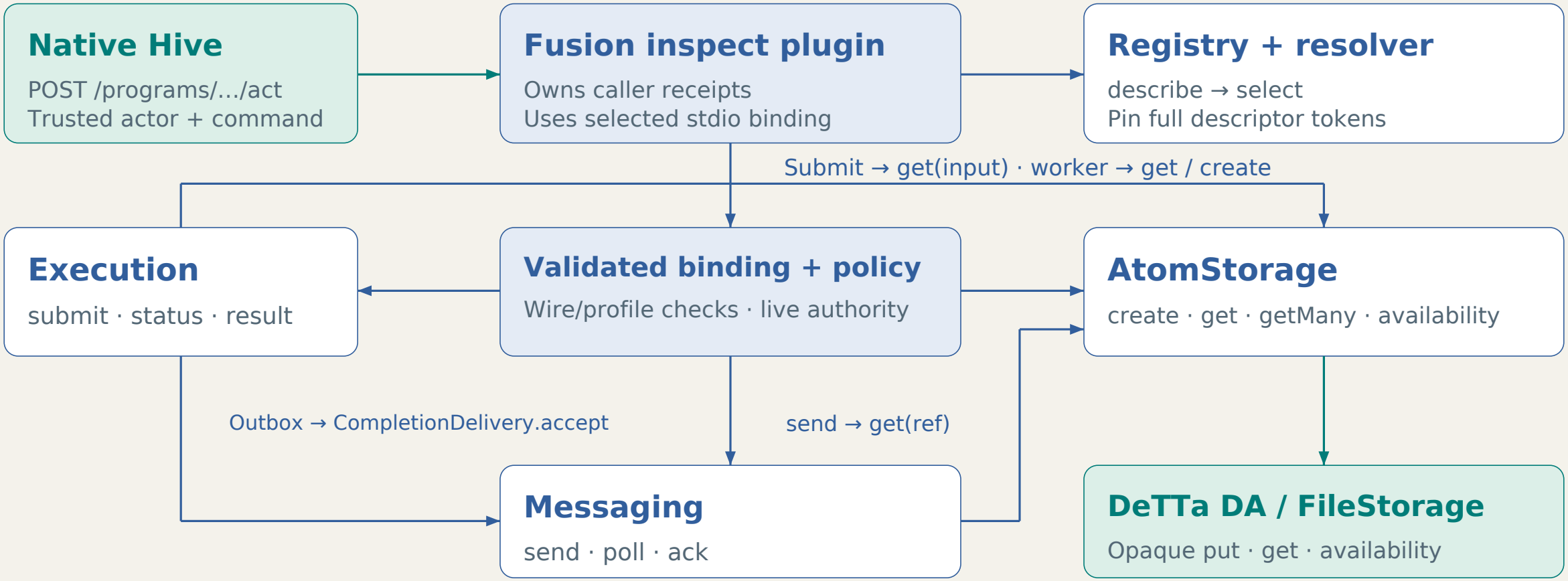
Function dependencies: the proposed cognitive path

Dashed arrows = specified calls. Solid arrow = implemented call. Direction = caller → provider.



Function dependencies: the real native path

Solid arrows = exercised calls. Every provider effect crosses an existing validated interface.



Optional finality uses a separate caller + approved digest + signer; M1.2 does not invoke it.

Separate owners now have separate durable commits

Result atoms, execution state, mailbox acceptance and caller acceptance do not share a transaction.

IMMUTABLE REFS + REPLAY

AtomStorage

Own metadata journal over opaque DA; independent file fixture also supported.

JOBS + COMPLETION OUTBOX

Execution

One terminal-state transaction; result persistence already committed.

INBOX + ACKNOWLEDGEMENTS

Messaging

Own acceptance and ack journal; delivery may repeat until acknowledged.

INTENT + ACCEPTED OUTCOME

Native caller

Minimal plugin receipt; stable job context survives loss of caller RAM.

MUTATION RESPONSE + FINGERPRINT

Wire admission

Colocated with execution; each effect owner also keeps replay identity.

CURRENT GRANTS + ROUTES

Authorization

Operator-owned configuration; access is rechecked on retries and work.

Co-location is allowed. Foreign SQL and shared transaction assumptions are not part of the contracts.

AtomStorage replacement is independently exercised

Protocol fixture callers exercise two providers and two transports; the worker logic is unchanged.

SELECTED ATOMSTORAGE	IN-PROCESS BINDING	PRIVATE STDIO BINDING
Real DeTTa libraries	PASS · independent workflow	PASS · independent workflow
Atomic-file fixture	PASS · independent workflow	PASS · independent workflow

Independent implementation

The file fixture does not inherit AtomOwner or use its SQLite atom table.

Fresh stores only

No accepted jobs or existing references are moved between providers.

Interface, implementation, transport and placement remain independent choices.

Compatibility binds semantics and selected replies

Consumer requirements are independent of provider advertisements.

Version
Wire remains 1.0.0

→

Profile
Exact behavioral semantics

→

Operations
Required methods and limits

→

Descriptor
Token checked on invocation and reply

AtomStorage
immutable-bytes/1

Messaging
durable-mailbox/2

Execution
inspect-atom/2

Descriptor implementation 0.1.1 • Deterministic inspect output algorithm 0.1.0

Descriptor hashes establish binding consistency; remote identity and provider truth remain separate.

Native identity and stable intent stay separate

Hive supplies the actor; the operator supplies the mapping; Fusion checks current authority.

```
action: inspect

params: {seed: 12}

idempotency_key: generated-intent-12


ctx.actor ← native authentication

ctx.command ← native request digest

credential ← private operator config
```

Only generated shareable input

The seed produces bytes; no private history or arbitrary remote operation is accepted.

Original input + original key

An ambiguous response replays the intent; it does not allocate a second job.

Authorization remains live

Even a recorded successful outcome must pass current module checks before return.

Scope labels, saved receipts and caller-supplied identity do not grant permission.

The intended Hive source now has a verified entrypoint

The supplied schematics identify the runtime and revision used for this engineering gate.

Schematics

4af00db4e395
Documentation points to Hive 34b0fb3.

Actual Hive source

34b0fb32ab38
Authorized SSH fetch; clean checkout.

Native baseline

34 selected tests passed
Upstream process driver is a fixture.

```
python -m hive → exclusive startup recovery → create_app

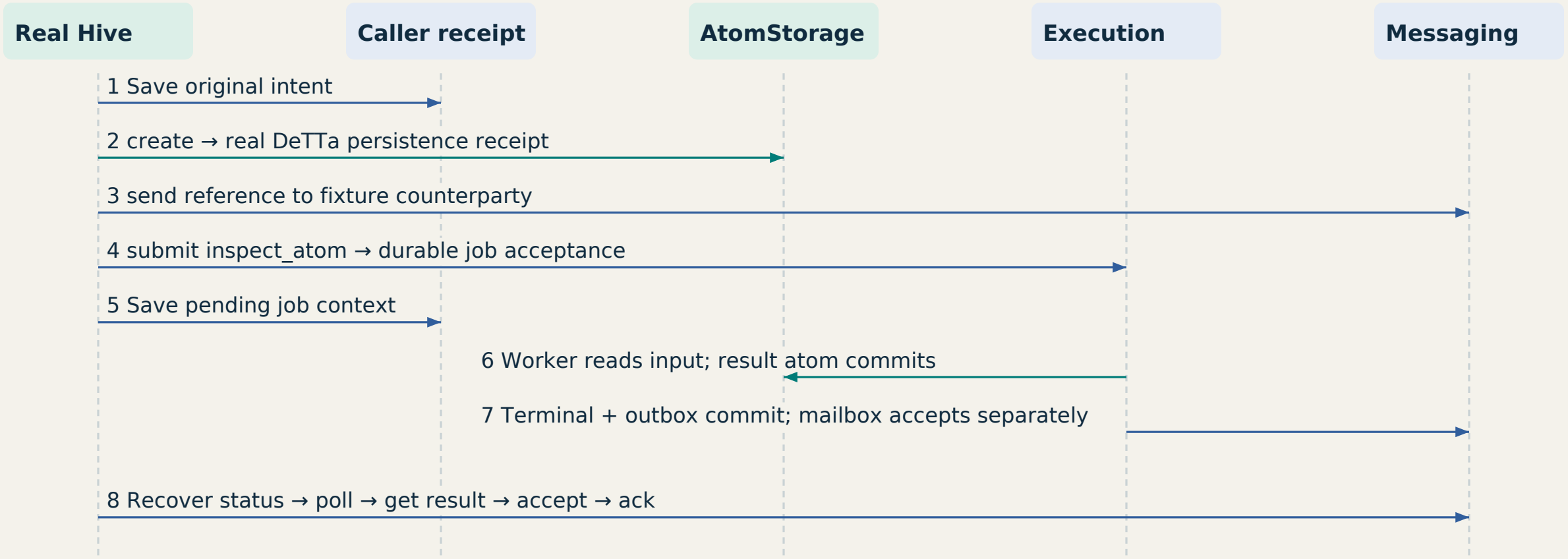
POST /api/programs/fusion-inspect/act

ProgramActions.person_act → Programs.act → plugin.act
```

Real native dispatch; ASGI TestClient is the transport. No fake Hive, live LLM or new server.

One native operation crosses the existing interfaces

The caller stores intent; each provider commits only its own state.



Six receipts describe six different facts

A pending native receipt can be reconciled; it is not a Hive goal or a claim of completed work.

1 · INPUT PERSISTENCE

Bytes have a scoped ref

Real DeTTa receipt; local files/process restart.

2 · JOB ACCEPTANCE

Work is durably queued

No claim yet about execution or reply delivery.

3 · RESULT + TERMINAL

Execution has an outcome

Atom result first; job terminal/outbox commit second.

4 · MAILBOX ACCEPTANCE

Completion is retained

Poll can repeat until the recipient acknowledges.

5 · CALLER ACCEPTANCE

One outcome is recorded

Known job, correlation and result bytes verified.

6 · ACKNOWLEDGEMENT

Delivery is acknowledged

Retry reuses the same key if the ack response is lost.

Native done = verified outcome + successful acknowledgement. Terminal failure stays an explicit error.

A restarted Hive caller recovers without its old RAM

Three kill points traverse the same authenticated native action route.

AFTER ADMISSION

Response lost

Fusion accepted the job; the caller has only its original intent. Same input + key recovers the same job.

SIGKILL → RESTART → REPLAY

AFTER SUBMISSION

Caller stops

The local receipt retains the job. Status /2 restores known correlation after a new process signs in.

SIGKILL → RESTART → REPLAY

AFTER ACCEPTANCE

Ack not yet sent

The outcome is durable. Redelivery is checked against it, then acknowledged without a second outcome row.

SIGKILL → RESTART → REPLAY

Observed: one job · one result atom · one completion · one accepted outcome identity

Known-job recovery uses durable context and status; unsolicited completions do not establish causality.

Worker failure and backpressure preserve pending work

A result can survive worker death; a full route can delay completion without losing it.

Result atom
Committed by AtomStorage

→

Job + outbox
Committed by execution

→

Completion
Accepted by Messaging

→

Caller outcome
Accepted, then acknowledged

REAL WORKER SIGKILL

Kill after result persistence

The job remains queued while its atom is durable. Restart reuses the logical result and later publishes completion.

NATIVE CALLER BACKPRESSURE

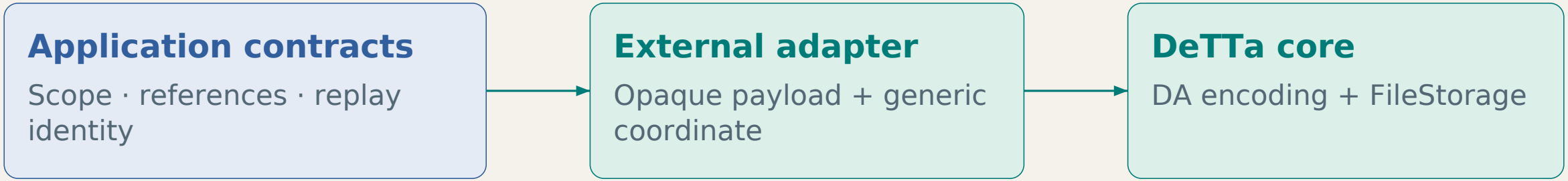
Full reply route

The outbox remains pending. The caller refuses an unrelated delivery; once capacity returns, the known completion is accepted.

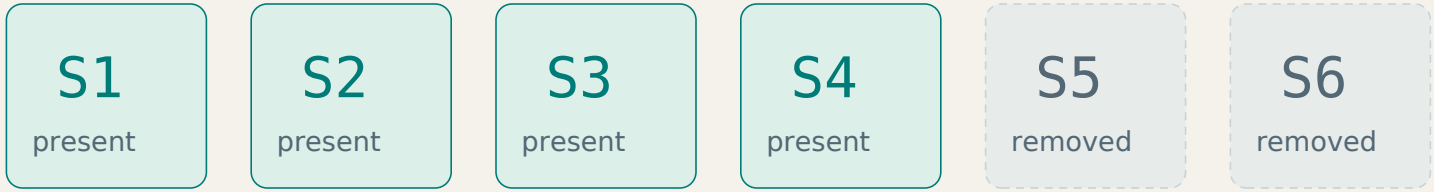
Worker revocation yields an explicit failed completion with no result atom; no silent success.

The DA boundary remains application-neutral

The external adapter translates application references into generic coordinates and opaque bytes.



4 data shares + 2 parity shares



Observed reconstruction

Cached payload + two shares removed: exact bytes recovered. A third missing share is unavailable.

The DA core does not interpret Hive identities, truth values, attention balances or job state.

Native-path refusals preserve authorization and replay

The host and module boundaries each keep their own checks.

ATTEMPT	OBSERVED RESULT	WHAT STAYS PROTECTED
Supply principal in action params	Native identity_in_request	Hive derives the actor
Map caller to the wrong scope	UNAUTHORIZED	Scope alone confers no access
Revoke caller after completion	UNAUTHORIZED on replay	Saved outcome grants no authority
Change profile / descriptor	UNSUPPORTED_PROFILE	No weaker provider fallback
Change input under original key	CONFLICT	No second job with new meaning
Revoke worker after admission	Explicit failed completion	No unauthorized result write

Real Hive authentication + development Fusion policy. Production isolation remains unproven.

Real runtime, real provider, fixtures and unrun scope

Each claim stays attached to the component and gate actually exercised.

COMPONENT	CLASSIFICATION	OBSERVED BOUNDARY
Hive + PeTTa	REAL NATIVE RUNTIME	Authenticated Programs dispatch / restart
DeTTa DA + FileStorage	REAL LOCAL PROVIDER	Persisted shares and reconstruction
Messaging / execution / worker	IMPLEMENTED LOCAL HARNESS	Separate owner commits; inspect_atom only
Counterparty / data / file provider	EXPLICIT FIXTURES	Generated sample and contract experiments
Fusion credential policy	DEVELOPMENT FIXTURE	Native auth is real; downstream policy local
Chain / cognitive implementations	NOT RUN IN M1.2	M1 chain historical; M2+ still specified

Fresh verification exercises distinct layers

Zero failures or skips in the final run. Fixture isolation is a subset, not extra unique coverage.

GATE / SUITE	PASSED	SCOPE OF THE OBSERVATION
Fixture-only isolation	51	Real DeTTa and Hive paths deliberately absent
Provider / transport matrix	90	Includes the 51 fixture tests; four workflows
Clean upstream DeTTa	62 + 38	DA + storage; historical dirty source had 40 storage tests
Native Hive baseline	34	Selected native tests; fixture process driver
Real-caller integration	13	Native dispatch, process recovery and refusals

Historical 27-test M1 results and earlier dirty sources are retained, not silently reused as current passes.

One trace ties native intent to acknowledged outcome

Identifiers below come directly from the recorded real Hive workflow.

Native caller

user:operator
Swarm s_2ccf3f15

→

Generated input

53 bytes · scope shared-test
a1b8c6b7-5dce-58c2…

→

Delegated job

inspect_atom / output 0.1.0
37fb2f12-9a0b-5eb7…

PERSISTED RESULT DIGEST

```
cd2f727689b41e243ee78207d058d00b
8be64932abbfd86a02851dfcbac400bb
```

Job ID = completion delivery ID; caller receipt = correlation ID.

Accepted and acknowledged

One durable outcome identity. Both atom receipts: real DeTTa, six shares / threshold four.

CHAIN = NOT_RUN for this result. The earlier M1 digest is a different, explicitly historical artifact.

Reproduction separates source, build and execution

Fresh disposable checkout; immutable authorized sources; locked new builds; actual runtime tests.

1 · SOURCE

Obtain exact bytes

Fusion commit/bundle; Hive and DeTTa via authorized SSH; public PeTTa and SWI commits.

2 · BUILD

Create new artifacts

Python 3.12.12, SWI 10.1.14 and Janus 1.5.3; locked packages; fresh native objects and venv.

3 · RUN

Exercise the seam

Both bridges and PeTTa evaluate; native baseline precedes the 13 actual caller cases.

Warm-cache assumptions are explicit

Python/uv and Cargo downloads reused; native pip install used --no-cache.

Clean DeTTa was rebaselined

Historical dirty bytes remain historical; fresh stores and all applicable tests were rerun.

Nine required gates passed; chain remains unrun

A fixture pass never substitutes for a real-provider or native-runtime pass.

BASELINE	PASS	Initial applicable suites executed
FIXTURE_REPRO	PASS	Fresh fixture-only run without real sources
HIVE_ENTRYPOINT	PASS	Real authentication and Programs dispatch
HIVE_RECOVERY	PASS	Restart, replay, completion and refusals
SCOPE	PASS	No M2, ECAN, inference control or migration
HIVE_SOURCE	PASS	Intended source and native entrypoint verified
DETTA_REPRO	PASS	Fresh clean provider build and conformance
HIVE_DETTA	PASS	Same workflow uses real DeTTa receipts
HIVE_REPRO	PASS	Declared fresh source/build recipe exercised
CHAIN	NOT_RUN	No same-artifact M1.2 chain submission

Historical chain reproduction is BLOCKED by missing exact F1R3 changes; empty-cache and full-Hive runs are NOT_RUN.

The real caller is proven; the cognitive increment is next

M1.2 establishes an integration seam. It does not select a numerical model or implement cognition.

M0 / M1

Contract slice

Module map, opaque storage, messaging and deterministic delegation.

M1.1

Owner boundaries

Independent AtomStorage replacement and separate durable commits.

M1.2

Real native caller

Verified Hive operation, durable caller recovery and fresh-source recipe.

M2+ · NOT STARTED

Cognitive work

Private episodes and reviewed revision; later attention and bounded inference.

Production boundaries

Isolation, authorization and distributed custody remain separate work.

Mathematical decisions

Revision model, lineage semantics, STI policy and specialists remain open.

State evolution

No live migration, populated-namespace switching or rolling upgrades.

Modularity now survives a real caller restart.

The native integration preserves the original boundaries.

01 State changes through its owner.

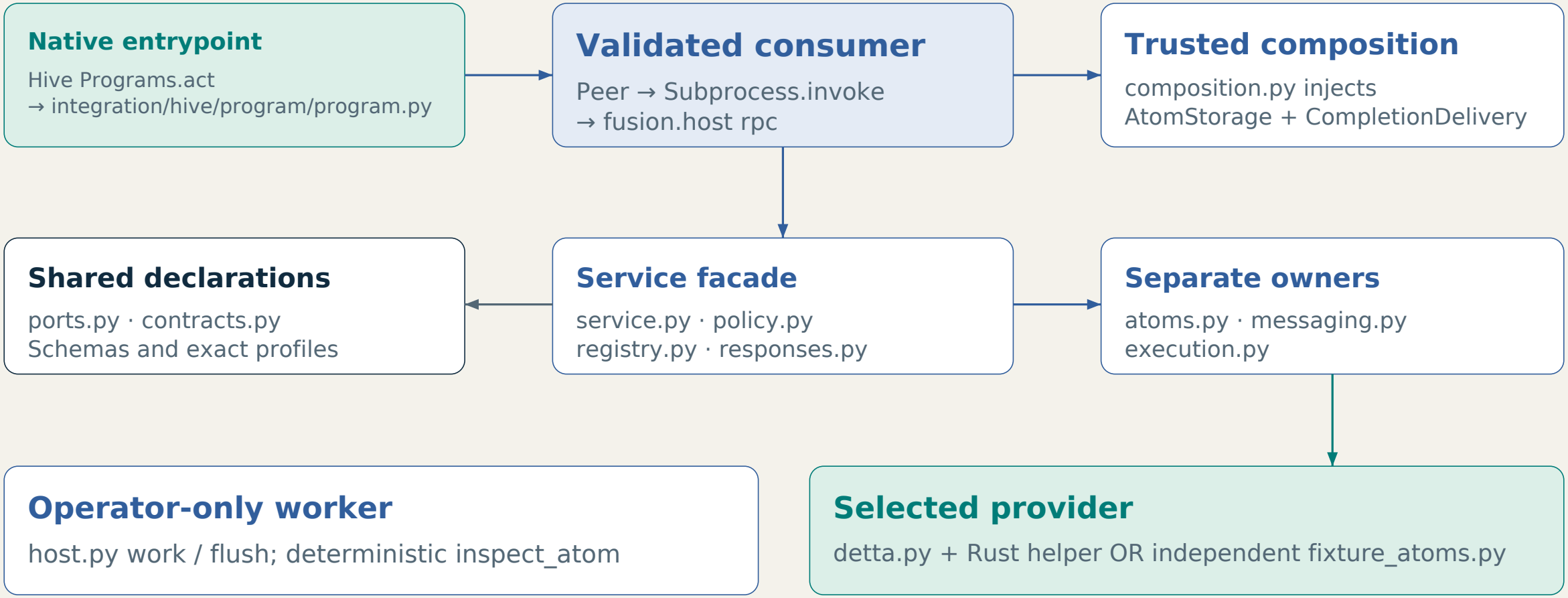
02 Providers remain replaceable by exact contract.

03 Each receipt names an observed, bounded guarantee.

Real Hive + real local DeTTa • Fixture counterparty • Chain NOT_RUN • Stop before M2

Implementation map: adapt at the native boundary

Native-specific translation stays outside the backend-independent modules.



Specified cognitive contracts carry test obligations

These are future acceptance cases, not executed M1.2 tests.

CONTRACT	SEMANTIC DECISION	NAMED CONFORMANCE OBLIGATIONS
Relations / Search	Ordered roles; dialect + watermark	REL-order / scope; SEARCH-watermark
Episodes · M2	Occurrence identity; sequence CAS	EP-order / duplicate / contrary / privacy
Belief revision · M2	Numerical model; lineage; independence	BR-overlap / cycle / contradiction / golden
Attention · M3	STI policy; logical time; budget; ties	ECAN-no-PLN / owner / budget / decay
PLN kernel · M3	Rule version; ordered premise snapshots	PLN-order / golden / purity / lineage
Inference control · M3	Candidate, rule, feedback, time limits	IC-bounds / feedback / replay / swap

Typed failures preserve the caller’s recovery choices

Unknown outcome is different from known rejection; an empty inbox is different from failure.

OBSERVATION / ERROR	MEANING	CALLER ACTION
NOT_FOUND	No such scoped atom metadata	Correct the reference
NOT_AVAILABLE	Known data / result temporarily unavailable	Retry or query the owner
OVERLOADED	Capacity or current writer lock unavailable	Back off; reuse key + semantic input
CONFLICT	Same key changed meaning / revision mismatch	Reconcile; do not blindly replay
UNAUTHORIZED / UNSUPPORTED_*	Access or contract requirements fail	Change authorization or requirements
Transport loss	Acceptance may already be durable	Recover by key / job; do not assume failure

Exact source identity includes the rebaseline

Commit prefixes are navigation aids; full SHAs, locks and clean-state checks are in the evidence.

INPUT	COMMIT / VERSION	USE AND SOURCE STATE
Fusion verification	bb22ef5e28b0	Tested implementation; clean checkout
Fusion evidence	4ee5cbe8b846	Documentation/evidence descendant; same code
omegahive-schematics	4af00db4e395	User-supplied runtime pointer
PhysixSNET/omegahive	34b0fb32ab38	Clean native source; actual dispatch
secured-contract-spaces	272888f264dd	Clean rebaseline; historical dirty bytes excluded
PeTTa	ae66fa8e41dc	Clean native language runtime
SWI / Python / Janus	10.1.14 / 3.12.12 / 1.5.3	Pinned native build; hashed Python lock

Reproduce the selected gates from a fresh checkout

Authorized source access and warm Cargo/Python downloads are prerequisites; see the runbook.

BUILD PINNED NATIVE DEPENDENCIES

```
uv run --python 3.11 --locked python \
scripts/build_hive.py artifacts/native
```

VERIFY FIXTURES, REAL DETTA AND NATIVE HIVE

```
uv run --python 3.11 --locked python \
scripts/verify_m12.py artifacts/native/environment.json
```

`Fixture-only: uv run --python 3.11 --locked python scripts/verify_m11.py --fixture-only`

Read manifest.json, verification.json, hive-trace.json and the preserved failure logs before extending scope.

Historical M1: one result reached an isolated chain

This is the earlier protocol-peer artifact. The M1.2 Hive result was not submitted.

PENDING
Signed deploy accepted

→

INCLUDED
Block observed; effects pending

→

FINALIZED
Deploy effects reported finalized

M1 recorded observation

Block height	**3**
Execution error	**false**
Rejection count	**0**
Execution cost	**351 phlo**

Historical commitment digest

```
f9236451ff785dc58315251cf864a3d1
0938412551cf45f448055061bc953ea1
```

One local validator; direct node observations.
Not a consensus proof or production test.

M1.2 CHAIN = NOT_RUN. Historical source reproduction requires exact F1R3 changes or a separate rebaseline.

Keep the guarantees inside their observed boundary

Local evidence is useful because the untested extensions remain explicit.

OBSERVED	UNRUN / UNSUPPORTED EXTENSION	WHY IT MATTERS
Same-host process restart	Power loss / distributed failover	Persistent files remained intact
One logical accepted outcome	Exactly-once physical execution	Deterministic work may repeat
Fresh-store provider swap	Live state / journal migration	Accepted refs and jobs are not moved
Exact descriptor/profile binding	Remote identity / verified proofs	A hash checks consistency, not truth
Fresh builds with declared caches	Entirely empty-cache build	Source access and downloads are prerequisites
One deterministic native operation	Full Hive suite / LLM / M2 cognition	Dispatch evidence is not cognitive capability